

IIT JAM 2012 Official Paper

Category: Classic Era (2005–2014) • Official Mock Test Series

TOTAL QUESTIONS

15

TOTAL MARKS

90 M

TEST DURATION

60 Min

PAPER PATTERN

MCQ

Section / Type		Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)		15	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2012 • Q1 • ID: JAM_2012_Q1)

MCQ

+6 M

Neg: -2.0

Q.1 Let $\{x_n\}$ be the sequence $+\sqrt{1}, -\sqrt{1}, +\sqrt{2}, -\sqrt{2}, +\sqrt{3}, -\sqrt{3}, +\sqrt{4}, -\sqrt{4}, \dots$.
If

$$y_n = \frac{x_1 + x_2 + \dots + x_n}{n} \text{ for all } n \in \mathbb{N},$$

then the sequence $\{y_n\}$ is

- (A) monotonic
- (B) NOT bounded
- (C) bounded but NOT convergent
- (D) convergent

Question 2 (JAM 2012 • Q2 • ID: JAM_2012_Q2)

MCQ

+6 M

Neg: -2.0

Q.2 The number of distinct real roots of the equation $x^9 + x^7 + x^5 + x^3 + x + 1 = 0$ is

- (A) 1 (B) 3 (C) 5 (D) 9

Question 3 (JAM 2012 • Q3 • ID: JAM_2012_Q3)

MCQ

+6 M

Neg: -2.0

Q.3 If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is defined by

$$f(x, y) = \begin{cases} \frac{x^3}{x^2 + y^4} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0), \end{cases}$$

then

- (A) $f_x(0, 0) = 0$ and $f_y(0, 0) = 0$
- (B) $f_x(0, 0) = 1$ and $f_y(0, 0) = 0$
- (C) $f_x(0, 0) = 0$ and $f_y(0, 0) = 1$
- (D) $f_x(0, 0) = 1$ and $f_y(0, 0) = 1$

Question 4 (JAM 2012 • Q4 • ID: JAM_2012_Q4)

MCQ

+6 M

Neg: -2.0

Q.4 The value of $\int_{z=0}^1 \int_{y=0}^z \int_{x=0}^y x y^2 z^3 dx dy dz$ is

- (A) $\frac{1}{90}$ (B) $\frac{1}{50}$ (C) $\frac{1}{45}$ (D) $\frac{1}{10}$

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Question 5 (JAM 2012 • Q5 • ID: JAM_2012_Q5)

MCQ

+6 M

Neg: -2.0

Q.5 The differential equation $(1 + x^2 y^3 + \alpha x^2 y^2) dx + (2 + x^3 y^2 + x^3 y) dy = 0$ is exact if α equals

- (A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) 2 (D) 3

Question 6 (JAM 2012 • Q6 • ID: JAM_2012_Q6)

MCQ

+6 M

Neg: -2.0

Q.6 An integrating factor for the differential equation $(2xy + 3x^2 y + 6y^3) dx + (x^2 + 6y^2) dy = 0$ is

- (A) x^3 (B) y^3 (C) e^{3x} (D) e^{3y}

Question 7 (JAM 2012 • Q7 • ID: JAM_2012_Q7)

MCQ

+6 M

Neg: -2.0

Q.7 For $c > 0$, if $a\hat{i} + b\hat{j} + c\hat{k}$ is the unit normal vector at $(1, 1, \sqrt{2})$ to the cone $z = \sqrt{x^2 + y^2}$, then

- (A) $a^2 + b^2 - c^2 = 0$ (B) $a^2 - b^2 + c^2 = 0$
(C) $-a^2 + b^2 + c^2 = 0$ (D) $a^2 + b^2 + c^2 = 0$

Question 8 (JAM 2012 • Q8 • ID: JAM_2012_Q8)

MCQ

+6 M

Neg: -2.0

Q.8 Consider the quotient group $\mathbb{Q}/\mathbb{Z}$ of the additive group of rational numbers. The order of the element $\frac{2}{3} + \mathbb{Z}$ in $\mathbb{Q}/\mathbb{Z}$ is

- (A) 2 (B) 3 (C) 5 (D) 6

Question 9 (JAM 2012 • Q9 • ID: JAM_2012_Q9)

MCQ

+6 M

Neg: -2.0

Q.9 Which one of the following is TRUE ?

- (A) The characteristic of the ring $6\mathbb{Z}$ is 6
(B) The ring $6\mathbb{Z}$ has a zero divisor
(C) The characteristic of the ring $(\mathbb{Z}/6\mathbb{Z}) \times 6\mathbb{Z}$ is zero
(D) The ring $6\mathbb{Z} \times 6\mathbb{Z}$ is an integral domain

Question 10 (JAM 2012 • Q10 • ID: JAM_2012_Q10)

MCQ

+6 M

Neg: -2.0

Q.10 Let W be a vector space over $\mathbb{R}$ and let $T: \mathbb{R}^6 \rightarrow W$ be a linear transformation such that $S = \{Te_2, Te_4, Te_6\}$ spans W . Which one of the following must be TRUE ?

- (A) S is a basis of W
(B) $T(\mathbb{R}^6) \neq W$
(C) $\{Te_1, Te_3, Te_5\}$ spans W
(D) $\ker(T)$ contains more than one element

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Question 11 (JAM 2012 • Q11 • ID: JAM_2012_Q11)

MCQ

+6 M

Neg: -2.0

Q.11 Consider the following subspace of $\mathbb{R}^3$:

$$W = \{(x, y, z) \in \mathbb{R}^3 \mid 2x + 2y + z = 0, 3x + 3y - 2z = 0, x + y - 3z = 0\}.$$

The dimension of W is

- (A) 0 (B) 1 (C) 2 (D) 3

Question 12 (JAM 2012 • Q12 • ID: JAM_2012_Q12)

MCQ

+6 M

Neg: -2.0

Q.12 Let P be a 4×4 matrix whose determinant is 10. The determinant of the matrix $-3P$ is

- (A) -810 (B) -30 (C) 30 (D) 810

Question 13 (JAM 2012 • Q13 • ID: JAM_2012_Q13)

MCQ

+6 M

Neg: -2.0

Q.13 If the power series $\sum_{n=0}^{\infty} a_n x^n$ converges for $x = 3$, then the series $\sum_{n=0}^{\infty} a_n x^n$

- (A) converges absolutely for $x = -2$
(B) converges but not absolutely for $x = -1$
(C) converges but not absolutely for $x = 1$
(D) diverges for $x = -2$

Question 14 (JAM 2012 • Q14 • ID: JAM_2012_Q14)

MCQ

+6 M

Neg: -2.0

Q.14 If $Y = \left\{ \frac{x}{1+|x|} \mid x \in \mathbb{R} \right\}$, then the set of all limit points of Y is

- (A) $(-1, 1)$ (B) $(-1, 1]$ (C) $[0, 1]$ (D) $[-1, 1]$

Question 15 (JAM 2012 • Q15 • ID: JAM_2012_Q15)

MCQ

+6 M

Neg: -2.0

Q.15 If C is a smooth curve in $\mathbb{R}^3$ from $(0,0,0)$ to $(2,1,-1)$, then the value of

$$\int_C (2xy + z)dx + (z + x^2)dy + (x + y)dz$$

is

(A) -1

(B) 0

(C) 1

(D) 2

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OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

IIT JAM 2012 Official Paper • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+6	D
2	MCQ	+6	A
3	MCQ	+6	B
4	MCQ	+6	A
5	MCQ	+6	B
6	MCQ	+6	C
7	MCQ	+6	A
8	MCQ	+6	B

Q. No.	Type	Marks	Official Key
9	MCQ	+6	C
10	MCQ	+6	D
11	MCQ	+6	B
12	MCQ	+6	D
13	MCQ	+6	A
14	MCQ	+6	D
15	MCQ	+6	C

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.